

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	28 June 2026
Roll No	23A51A4229
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

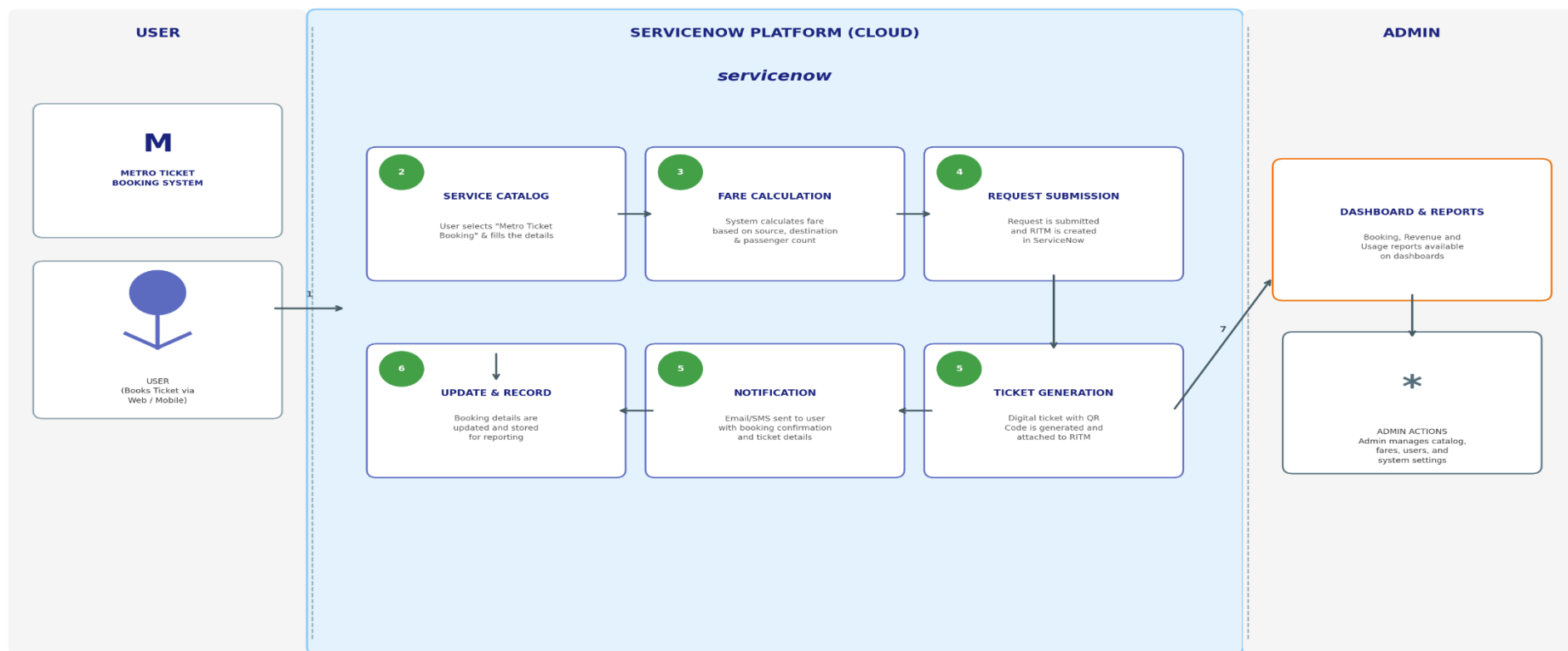


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interface for passengers to book metro tickets online	ServiceNow Service Catalog, Portal UI
2.	Application Logic-1	Fare calculation based on source and destination stations	Catalog Client Scripts (JavaScript)
3.	Application Logic-2	Server-side data retrieval and validation	Script Includes, GlideAjax
4.	Application Logic-3	Automated ticket booking workflow and notifications	Flow Designer, Business Rules
5.	Database	Stores metro stations, fares, and booking information	ServiceNow Tables (u_metro_station, u_metro_fare, sc_req_item)
6.	Cloud Database	Cloud-based data storage managed by ServiceNow	ServiceNow Cloud Database
7.	File Storage	Stores QR codes, ticket attachments, and records	ServiceNow Attachments
8.	External API-1	QR Code generation for digital tickets	QR Code Generator API / Scripted QR Generation
9.	External API-2	Email notification service for ticket confirmation	ServiceNow Notification Engine
10.	Machine Learning Model	Ticket booking trends, revenue, and usage reports	ServiceNow Reports & Dashboards
11.	Infrastructure (Server / Cloud)	Application hosting and execution environment	ServiceNow SaaS Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open Source Frameworks	Client-side scripting and customization used for fare calculation and form validation	JavaScript, GlideAjax
2.	Security Implementations	Role-based access control for admins and users; authenticated access through ServiceNow portals; secure data storage and transmission	ServiceNow ACLs, User Roles, HTTPS, IAM Console
3.	Scalable Architecture	ServiceNow cloud-based multi-tenant architecture supports growing users and ticket bookings without infrastructure changes	ServiceNow SaaS Architecture, Multi-Tenant Cloud Platform
4.	Availability	High availability through ServiceNow cloud infrastructure with built-in redundancy and managed services	ServiceNow Cloud Infrastructure, Redundant Data Centers
5.	Performance	Fast ticket booking through server-side processing, optimized database queries, and cloud-hosted services	ServiceNow Script Includes, Optimized GlideRecord Queries

References:

<https://www.servicenow.com/docs/en/servicenow-platform/service-catalog/service-catalog.html>

<https://www.servicenow.com/docs/en/application-development/building-applications/flow-designer.html>

https://www.servicenow.com/docs/en/api-reference/c_GlideAjaxAPI.html

<https://developer.servicenow.com>

<https://www.servicenow.com/community/>